

ASYMPTOTICS OF PARTITION FUNCTIONS OF COULOMB GASES

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Supervisor: Guilherme Silva



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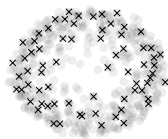
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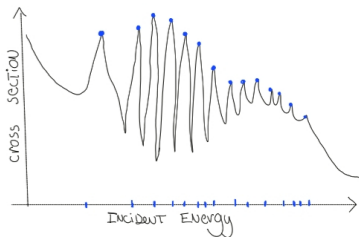
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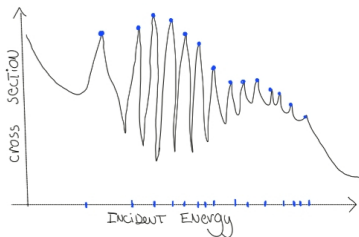


RANDOM MATRICES



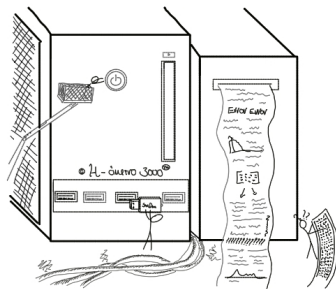
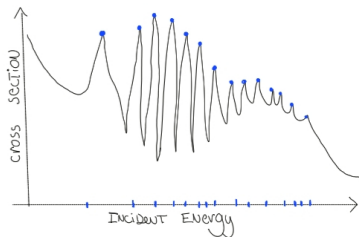


$$\mathcal{H}\phi_i = E_i\phi_i$$





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EUGENE WIGNER - *On the Wigner Surmise* (1956)

"Perhaps I am now too courageous when I try to guess the distribution of the distances between successive levels (of energies of heavy nuclei) [...] The question is simply what are the distances of the characteristic values of a symmetric matrix with random coefficients."

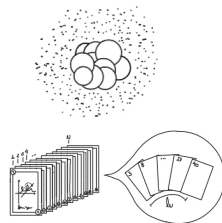
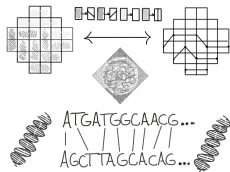
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A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} \\ x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} \\ x_{3,1} & x_{3,2} & x_{3,3} & x_{3,4} \\ x_{4,1} & x_{4,2} & x_{4,3} & x_{4,4} \\ x_{5,1} & x_{5,2} & x_{5,3} & x_{5,4} \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
Discrete	$\mathbb{R}$	General
Uniform	$\mathbb{C}$	Hermitian
Gaussian	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$

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A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -1 & -1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
Discrete	$\mathbb{R}$	General
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$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n p(x_{i,j}) = U^{n^2}(\{0,1\})$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} 0.213 & 0.368 & 0.912 & 0.881 \\ 0.759 & 0.692 & 0.000 & 0.332 \\ 0.898 & 0.987 & 0.422 & 0.623 \\ 0.145 & 0.880 & 0.522 & 0.686 \\ 0.753 & 0.687 & 0.787 & 0.515 \end{bmatrix}$$

$p(x_{i,j})$	$\mathbb{S}$	$\mathcal{M}_n$
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A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -0.022 & -1.340 & -0.483 & -1.134 \\ 1.752 & -1.644 & 1.081 & -0.936 \\ -1.058 & -0.091 & 0.544 & -0.382 \\ 0.633 & 0.567 & 0.606 & -0.621 \\ -0.316 & -1.128 & -0.150 & 0.251 \end{bmatrix}$$

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Ginibre Orthogonal Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n e^{-x_{i,j}^2}$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -2.0+0.1i & -0.2-0.9i & -0.4-0.3i & -0.3+1.0i \\ -1.6-0.8i & 0.3-0.6i & -1.7+0.9i & -0.1+0.3i \\ -0.5-0.9i & -0.5+0.2i & 1.2+1.0i & 0.4-1.3i \\ -0.6+2.4i & 0.1-0.7i & -0.2+1.6i & 0.6-0.4i \\ 0.1-1.3i & -0.2-0.9i & 1.2-1.6i & 2.1+2.0i \end{bmatrix}$$

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Ginibre Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i,j=1}^n e^{-x_{i,j}^2} e^{-y_{i,j}^2}$$

A *random matrix* is a matrix with *random variables* as entries.

$$\begin{bmatrix} -0.8 & -0.2-1.5i & 0.9-0.6i & -3.2+2.3i \\ -0.2+1.5i & 0.5 & -1.8-0.8i & 1.5+1.8i \\ 0.9+0.6i & -1.8+0.8i & 1.8 & 2.1+2.5i \\ -3.2-2.3i & 1.5-1.8i & 2.1-2.5i & 2.4 \end{bmatrix}$$

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Gaussian Unitary Ensemble

$$\mathcal{P}(\hat{X}) = \prod_{i=1}^n e^{\frac{z_{i,j}^2}{2}} \prod_{i>j}^n e^{z_{i,j}^2}$$

We can think of the matrix itself as a random variable, a measurable map from the a probability measure space to $\mathcal{M}_n$.

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$\mathcal{P}(\hat{X})$	$\mathbb{S}$	$\mathcal{M}_n$
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$e^{-n \text{Tr}\{(\hat{X}^* \hat{X})\}}$	$\mathbb{C}$	Hermitian
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We can think of the matrix itself as a random variable, a measurable map from the a probability measure space to $\mathcal{M}_n$.

Normal matrices are similar to diagonal matrices, i.e.

$$\hat{X} = \hat{U} \hat{\Lambda} \hat{U}^*.$$

Moreover, the spectra is typically simple.

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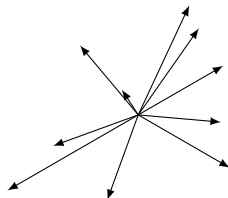
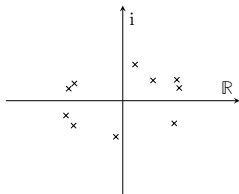
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Invariance by unitary conjugation means independence of coordinates.

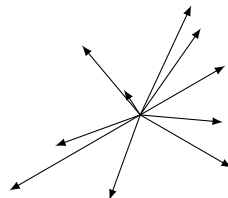
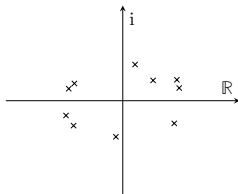
$$d\mathcal{P}(\hat{X}_1) = d\mathcal{P}(\hat{X}_2) \iff \hat{X}_1 = \hat{U}\hat{X}_2\hat{U}^*$$

for some unitary U .

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We want that $d\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})d\hat{X}$ is invariant.

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$$\exp\left(-n\text{tr}\left\{Q(\hat{U}\hat{X}_1\hat{U}^*)\right\}\right) = \exp\left(-n\text{tr}\left\{Q(\hat{U}^*\hat{U}\hat{\Lambda})\right\}\right) = \exp\left(-n\text{tr}\left\{Q(\hat{X}_1)\right\}\right)$$

$$\exp\left(-n\text{tr}\left\{Q(\hat{U}\hat{\Lambda}\hat{U}^*)\right\}\right) = \exp\left(-n\sum_i^n Q(\lambda_i)\right).$$

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$$d\hat{X} = \prod_{1 \leq i < j \leq n} |z_i - z_j|^2 d\hat{U} \prod_{i=1}^n d^2 z_i,$$

where $d\hat{U}$ is the Haar measure on eigenvectors and $d^2 z$ is Lebesgue.

We set the ensemble

$$\mathcal{E} = (\mathcal{S}, \mathcal{P}) \quad \text{with} \quad \begin{cases} \mathcal{S} = \mathcal{N}_n(\mathbb{C}) \\ \mathcal{P}(\hat{X}) = \exp \left\{ -n \operatorname{Tr} \left(Q(\hat{X}) \right) \right\} \end{cases}$$

such that

$$d\mathcal{P}(\hat{X}) = \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)} d^2 z_j d\hat{U}.$$

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$$\begin{aligned} \int_{\mathcal{M}_n} d\mathcal{P}(\hat{X}) &= \int_{\mathbb{C}^n} \int_{\tilde{U}(n)} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)} d^2 z_j d\hat{U} \\ &= \int_{\mathbb{C}^n} \frac{\prod_{i < j}^n |z_i - z_j|^2 \prod_{j=1}^n e^{-nQ(z_j)}}{\mathcal{Z}_n^Q} d^2 z_j \end{aligned}$$

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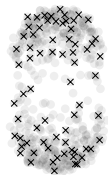
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we have

$$p_n(\{\lambda_i\}_{i=1}^n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n \mathrm{e}^{-nQ(z_i)}$$



MODELING OUR MODEL

We know that

$$p_n(\{\lambda_i\}_{i=1}^n) = p\left(\begin{array}{c} \text{Diagram of } n \text{ points in } \mathbb{R}^2 \\ \text{with axes } | \cdot |_1 \text{ and } \mathbb{R} \end{array}\right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)}$$

We know that

$$p_n(\{\lambda_i\}_{i=1}^n) = p \left(\begin{array}{c} \text{Diagram of } n \text{ points in } \mathbb{R}^d \\ \text{with axes } \mathbb{R} \text{ and } \mathbb{I} \end{array} \right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)}$$

We can notice two effects, one coming from the Jacobian and other from Q .

$$p_n \left(\begin{array}{c} \text{Diagram of } n \text{ points in } \mathbb{R}^d \\ \text{with axes } \mathbb{R} \text{ and } \mathbb{I} \end{array} \right) > \left\{ \begin{array}{l} \text{ } \end{array} \right.$$

We know that

$$p_n(\{\lambda_i\}_{i=1}^n) = p \left(\text{Diagram of points in a circle} \right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)}$$

We can notice two effects, one coming from the Jacobian and other from Q .

$$p_n \left(\text{Diagram of points in a circle} \right) > \left\{ p_n \left(\text{Diagram of points in a circle} \right), \text{ the interactions dominates.} \right.$$

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$$p_n \left(\text{plot of points in a circle} \right) > \begin{cases} p_n \left(\text{plot of points clustered near origin} \right), & \text{the interactions dominates.} \\ p_n \left(\text{plot of points scattered in a larger region} \right), & \text{the external potential dominates.*} \end{cases}$$

We know that

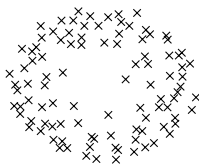
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We know that

$$\begin{aligned} p_n(\{\lambda_i\}_{i=1}^n) &= p\left(\text{Diagram of } n \text{ points in the plane}\right) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i<j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)} \\ &= \frac{1}{\mathcal{Z}_n^Q} \exp\left\{-n^2 \left(\frac{1}{n} \sum_{i=1}^n Q(z_i) + \frac{1}{n^2} \sum_{i \neq j} \ln |z_i - z_j|\right)\right\} \end{aligned}$$

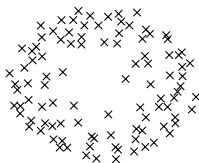
POINT PROCESS

Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$.



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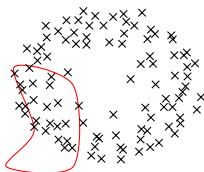


Define a quantity $\xi(A)$ s.t.

$$\xi(A) = \sum_{k=1}^n \delta(z_k)$$

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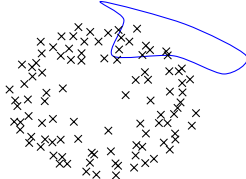


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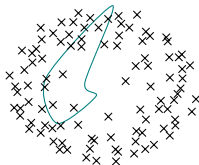


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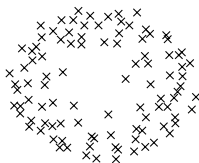


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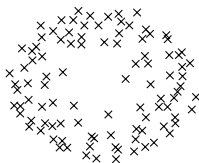
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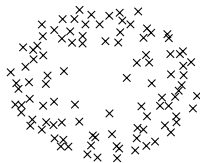


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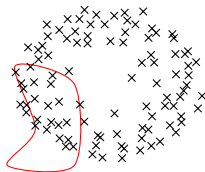
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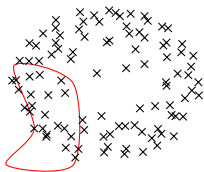


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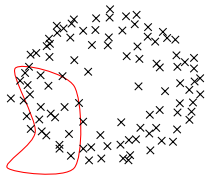
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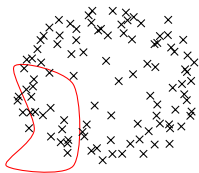
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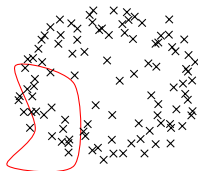
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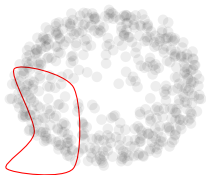


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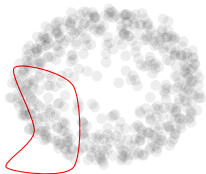
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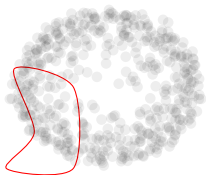
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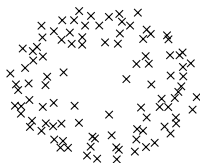
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where d^2z is the area measure. The **1-point correlation function** p_1 represents the infinitesimal probability of a point of the process being at a location z , i.e.

$$\mathbb{P}(\xi(d^2z) = 1) = p_1(z) d^2z.$$

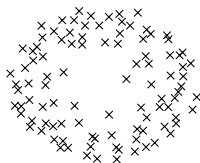
VARIATIONAL PROBLEM



$$p\left(\begin{array}{c} \text{\tiny $\bullet$}\\ \vdots \\ \text{\tiny $\bullet$}\end{array}\right)=\frac{{e^{-n^2 {\cal H}({\chi}_n)}}}{{\cal Z}_n^Q},$$

$$\mathcal{H}(\chi_n) = \frac{1}{n^2} \sum_{i \neq j}^n \ln |z_i - z_j| + \frac{1}{n} \sum_{i=1}^n Q(z_i).$$

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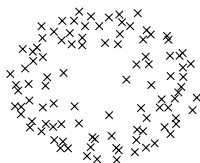


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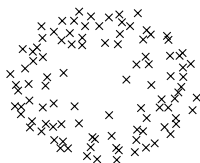


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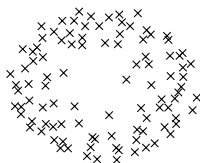


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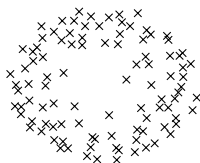


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Associated with the variational problem defined by

$$E^w = \inf_{\mu \in \mathcal{M}^1} \left[\int \int_{t \neq s} \ln |t - s|^{-1} d\mu(t) d\mu(s) + \int Q(s) d\mu(s) \right] = I_Q(\mu_Q),$$

one defines the so-called *Euler-Lagrange* type variational equations.

THEOREM

(Strong Variational Equations) Suppose that $d\mu_Q = \mu(z)d^2z$ for some continuous function μ of compact support. The equilibrium measure μ_Q for the given variational problem, where $\Sigma = \mathbb{C}$ satisfies the following conditions. There exists a real constant l such that

①

$$2 \int \ln |z - t|^{-1} d\mu_Q(t) + Q(z) \geq l \quad (2.1)$$

for all $z \in \mathbb{C}$.

②

$$2 \int \ln |z - t|^{-1} d\mu_Q(t) + Q(z) = l \quad (2.2)$$

on $\{z : \mu(z) > 0\}$.

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- 3 Furthermore, assume Q to be smooth, sub-harmonic in $\mathbb{C}$, and strictly sub-harmonic in a neighborhood of the $\text{supp}(\mu_Q)$.

THEOREM

If r_0 is the largest solution of $rq'(r) = 0$ and r_1 is the smallest solution of $rq'(r) = 2$ then the support of μ_Q is given by the ring $S = \{z | r_0 \leq |z| \leq r_1\}$ and $d\mu_Q$ is given by

$$d\mu_Q(z) = \frac{1}{4\pi} (rq'(r))' \delta_S dr d\phi, \quad z = r e^{i\phi}. \quad (2.3)$$

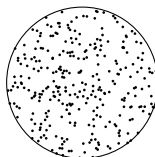
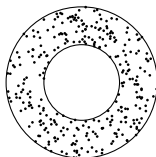
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We call **Droplet** the support of μ_Q , which in our case is the set of all complex values z such that $r_0 \leq |z| \leq r_1$.

Let $\mu \in \mathcal{M}^1(\mathbb{C})$ be a measure such that $d\mu = \frac{1}{4\pi}(rq'(r))'\delta_S d\phi dr$.

$$\int_{\mathbb{C}} d\mu = 1, \quad (2.4)$$

that is,

$$r_1 q'(r_1) - r_0 q'(r_0) = 2. \quad (2.5)$$

but r_0 is the largest solution of $rq'(r) = 0$ and r_1 is the smallest solution of $rq'(r) = 2$. Now, by Euler-Lagrange Theorem, if μ satisfies that

$$2U^\mu(z) \geq I - Q(z), \quad z \in \mathbb{C} \quad (2.6)$$

$$2U^\mu(z) = I - Q(z), \quad z \in \{z \mid r_0 \leq |z| \leq r_1\}. \quad (2.7)$$

with $I_Q(\mu) < \infty$, then $\mu_Q = \mu$. We use the formula

$$\int_{\pi}^{-\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi = \begin{cases} 2\pi \ln \frac{1}{r}, & |z| \leq r, \\ 2\pi \ln \frac{1}{|z|}, & |z| > r. \end{cases} \quad (2.8)$$

With this, we calculate that

$$U^\mu(z) = \frac{1}{4\pi} \int_{r_0}^{r_1} (rq'(r))' \int_{-\pi}^{\pi} \ln \frac{1}{|z - r e^{i\phi}|} d\phi dr.$$

And, for $|z| < r_0$

$$U^\mu(z) = \frac{1}{2} \int_{r_0}^{r_1} (rq'(r))' \ln \left(\frac{1}{r} \right) dr = -\frac{1}{2} (2 \ln(r_1) - q(r_1)) - \frac{1}{2} q(r_0),$$

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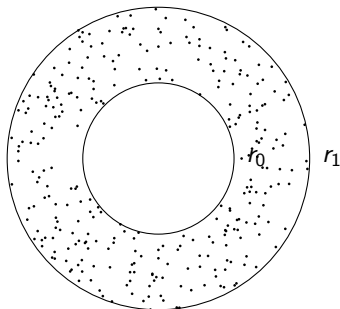
$$U^\mu(z) = \frac{1}{2} \int_{r_0}^{|z|} (rq'(r))' \ln \left(\frac{1}{r} \right) dr + \frac{1}{2} \int_{|z|}^{r_1} (rq'(r))' \ln \left(\frac{1}{z} \right) dr, \quad (2.9)$$

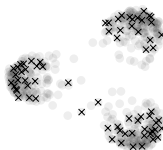
$$= -\frac{1}{2} (2 \ln(r_1) - q(r_1)) - \frac{1}{2} q(|z|). \quad (2.10)$$

Define $l = -(2\ln(r_1) - q(r_1))$, then

$$2U^\mu(z) = \begin{cases} l - q(r_0), & |z| < r_0, \\ 2\ln \frac{1}{|z|}, & |z| > r_1, \\ l - q(|z|), & r_0 \leq |z| \leq r_1. \end{cases} \quad (2.11)$$

We need to prove that for $z \in \mathbb{C}$, we have $2U^\mu(z) \geq l - q(|z|)$ while on $r_0 \leq |z| \leq r_1$ we have $2U^\mu(z) = l - q(|z|)$.





ASYMPTOTICS ON THE PARTITION FUNCTION

QUICK RECAP

$$d\mathcal{P}(\hat{X}) = \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_i)} d^2 z_i d\hat{U}.$$

$\mathcal{P}(\hat{X})$	$\mathbb{S}$	$\mathcal{M}_n$
$e^{-n \operatorname{Tr}(\hat{M}^2)}$	$\mathbb{R}$	General
$e^{-n \operatorname{Tr}\{(\hat{X}^* \hat{X})\}}$	$\mathbb{C}$	Hermitian
$e^{-n \operatorname{Tr}\{Q(\hat{X})\}}$	$\mathbb{H}$	Normal
$\vdots$	$\vdots$	$\vdots$



Orthogonal Polynomial Ensemble

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$$p_n(\{\lambda_i\}_{i=1}^n) = \frac{1}{Z_n^Q} \prod_{i < j}^n |z_i - z_j|^2 \prod_{j=i}^n e^{-nQ(z_j)}$$

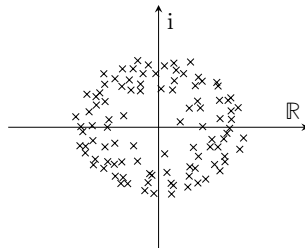
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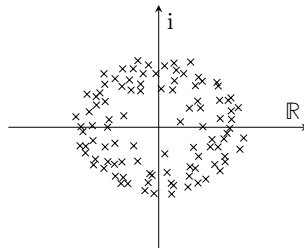
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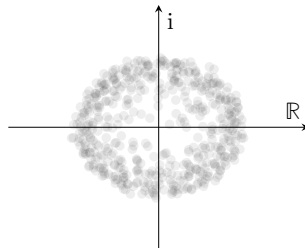
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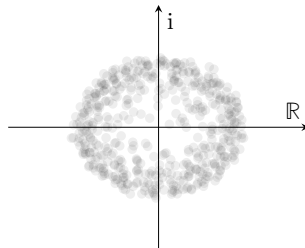


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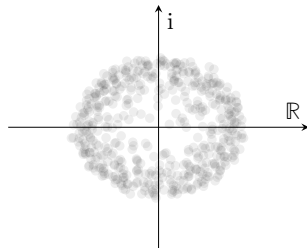


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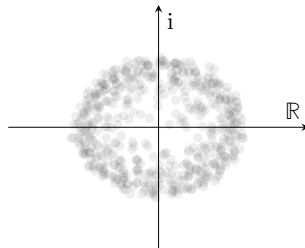
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$$\ln(\mathcal{Z}_n^Q) = -I_Q n^2 + \frac{1}{2} n \ln n + E_Q n + G \ln n + F^Q + o(1).$$

SKETCH

- ① *Partition functions can be expressed in terms of the **norming constants**: We can write the free energy as*

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

- ② *Asymptotics of norming constant using the asymptotic **Laplace technique**.*

$$h_j = \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2z \approx \frac{e^{-nV_{\tau(j)}(r_c(\tau(j)))}}{\sqrt{n}}.$$

- ③ *Asymptotic of the summation on $\ln h_j$ using **quadrature techniques**:*

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- ① *Partition functions can be expressed in terms of the **norming constants**: We can write the free energy as*

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Remember that the j.p.d.f. considered, with $w(z_i) = e^{-nQ(z_i)}$, is

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THEOREM

Let $\{q_{j,n}\}_{j=0}^\infty$ be a family of monic polynomials such that each $q_j(\lambda)$ is of degree j . Impose also that exists finite positive constants $\{h_{j,n}\}_{j=0}^\infty$ s.t.

$$\langle q_{j,n}, q_{k,n} \rangle_n = \int_{\mathbb{C}} q_{j,n}(z) \overline{q_{k,n}(z)} e^{-nQ(z)} d^2 z = h_{j,n} \delta_{\{j=k\}}$$

for any $j, k > 0$. Then, $K_n(x, y)$ has reproducing properties, i.e.

$$\int_{\mathbb{C}} \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^{k+1} d^2 \lambda_{k+1} = (n - k) \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^k.$$

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COROLLARY

Because of

$$\int_{\mathbb{C}} \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^{k+1} d^2 \lambda_{k+1} = (n - k) \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^k,$$

the partition function $\mathcal{Z}_n^Q$ can be written as

$$\mathcal{Z}_n^Q = \prod_{l=1}^n h_{n,l} \int_{\mathbb{C}^n} \det [K_n(\lambda_i, \lambda_j)]_{i,j=1}^n \prod_{i=1}^n d^2 \lambda_i = n! \prod_{l=1}^n h_{n,l}$$

NORMING CONSTANTS

Defining $z = r e^{i\theta}$ ($|z| = r$),

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We redefine our potential using

$$V_{\tau(j)}(r) = q(r) - 2\tau(j) \ln(r),$$

where $\tau(j) := j/n \in [0, 1)$. Then, rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r e^{-nV_{\tau(j)}(r)} dr.$$

Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V'_\tau(r) = q'(r) - \frac{2\tau}{r} \quad \implies \quad r_c q'(r_c) = 2\tau.$$

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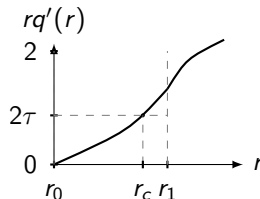
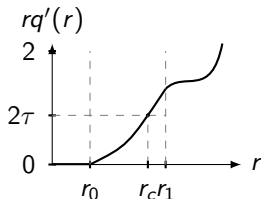
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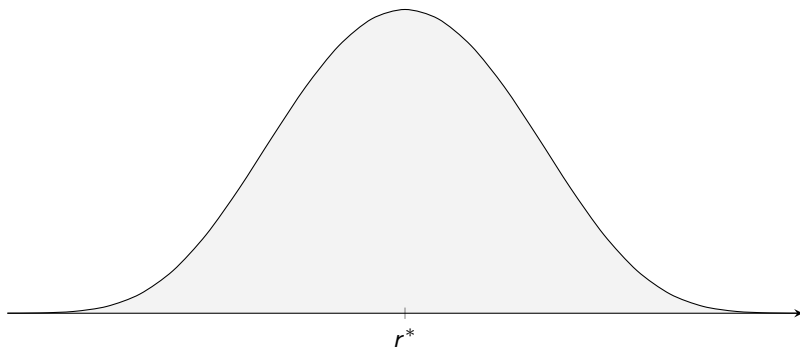
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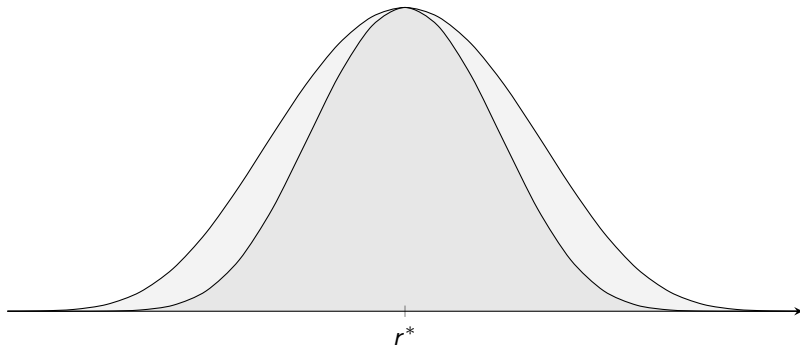
$$\begin{cases} r_0, & \text{the largest solution of } rq'(r) = 0, \\ r_1, & \text{the smallest solution of } rq'(r) = 2. \end{cases}$$



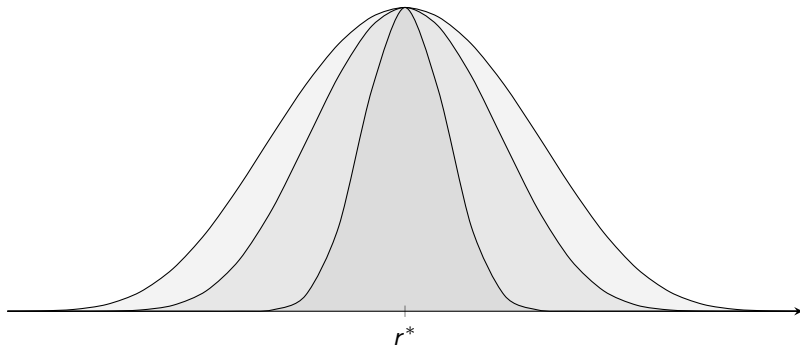
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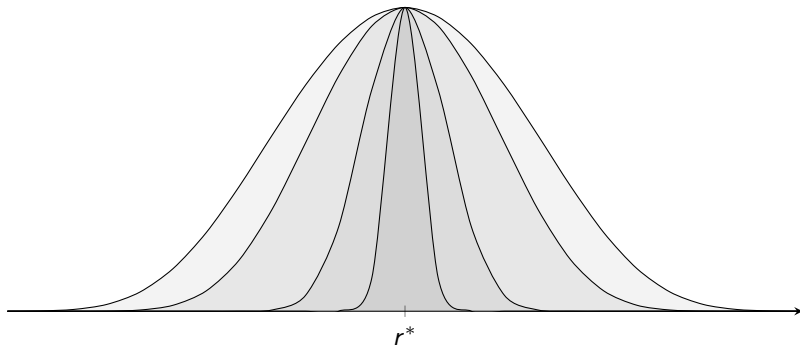
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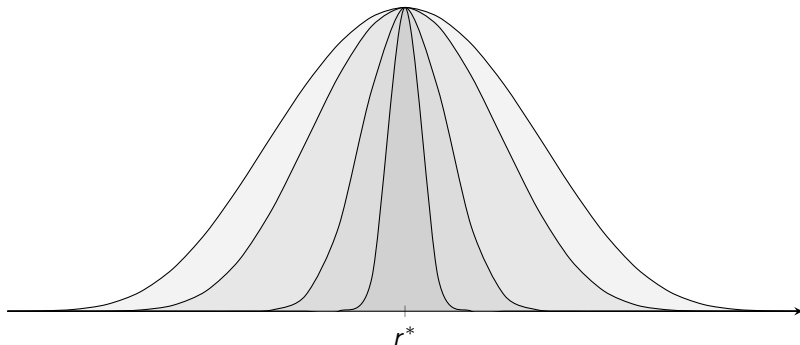
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$$\lim_{\lambda \rightarrow \infty} \frac{e^{-\lambda(f(r^*))}}{e^{-\lambda(f(r^*-\epsilon))}} = \lim_{\lambda \rightarrow \infty} e^{-\lambda(f(r^*)-f(r^*-\epsilon))} = \lim_{\lambda \rightarrow \infty} e^{\lambda|c|} \rightarrow +\infty. \quad (3.1)$$

LAPLACE TECHNIQUE

We start by approximating $V_{\tau(j)}$ around its global maximum by Taylor's theorem

$$V_{\tau(j)}(r) = V_{\tau(j)}(r^*) - \frac{1}{2}|V_{\tau(j)}^{(2)}(r^*)|(r - r^*)^2 + \epsilon((r - r^*)^3)$$

and then perform a change of variables $\frac{r}{\sqrt{n}} = s - r^*$, so that

$$\int_{r^* - \frac{\ln(n)}{\sqrt{n}}}^{r^* + \frac{\ln(n)}{\sqrt{n}}} 2s e^{-nV_{\tau(j)}(s)} ds = \frac{e^{-nV_{\tau(j)}(r^*)}}{\sqrt{n}} \int_{-\ln(n)}^{\ln(n)} 2 \left(r^* + \frac{r}{\sqrt{n}} \right) e^{\frac{1}{2}|V_{\tau(j)}^{(2)}(r^*)|r^2 - \epsilon\left(\frac{r^3}{\sqrt{n}}\right)} dr.$$

and

$$\int_{x^* + \frac{\ln(n)}{\sqrt{n}}}^{\infty} |2s| e^{-nV_{\tau(j)}(s)} d(s) \leq e^{-nV_{\tau(j)}(r^* + \ln(n)/\sqrt{n})} \|g\|_{L^1}$$

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THEOREM

(Laplace's Method) Let Φ be

$$\Phi(\lambda) := \int_a^b g(t) e^{-\lambda\phi(t)} dt \quad (3.2)$$

with a finite and $g \in L^1(a, b)$. Assume that ϕ has a unique global minimum at $t = c \in (a, b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form $g(t) = (t - a)^\alpha g_0(t)$ for $|t - a| < \delta$ and g_0 continuously differentiable in $[c - \delta, c + \delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c - \delta, c + \delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = e^{-\lambda\phi(c)} \left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + \mathcal{O}(\lambda^{-3/2}) \right)$$

as $\lambda \rightarrow \infty$.

THEOREM

We have the following asymptotic of n for h_j , with $0 \leq j \leq n-1$,

$$h_j = \frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right],$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2} \frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3} \frac{5}{24}.$$

QUADRATURE

We have

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

where

$$h_j = \int_{\mathbb{C}} |z|^{2j} e^{-nQ(z)} d^2 z \approx \frac{e^{-nV_{\tau(j)}(r_c(\tau(j)))}}{\sqrt{n}}.$$

QUADRATURE

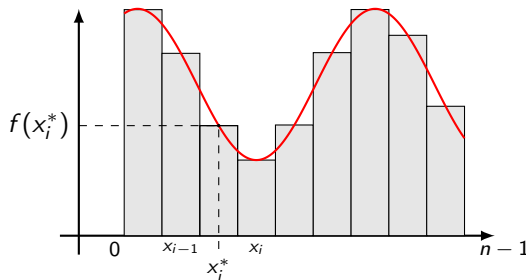
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$$\Delta_n \sum_{j=0}^{n-1} \ln h_j \approx \int_a^b \ln h_j$$



QUADRATURE I

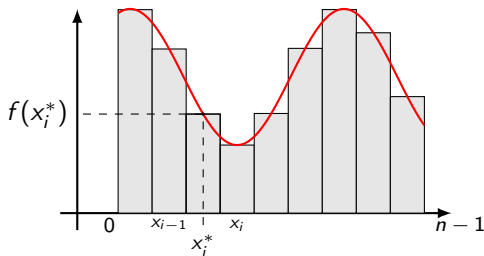
We begin with $h = (b - a)/n$, s.t.

$$\begin{aligned}
 \int_a^b f(x) dx &\approx h \sum_{j=0}^n f(x_j^*), \\
 &= h \sum_{j=0}^n \frac{f(x_i) + f(x_{i-1})}{2}, \\
 &= h \left[\frac{f(a)}{2} + f(a+h) + \cdots + f(b-h) + \frac{f(b)}{2} \right], \\
 &= h \sum_{j=1}^{n-1} f(x_j^*) + \frac{h}{2} (f(b) + f(a)), \\
 &= h \sum_{j=0}^{n-1} f(x_j^*) + \frac{h}{2} (f(b) - f(a)).
 \end{aligned}$$

THEOREM

(Euler-Maclaurin Formula) If m and n are natural numbers and $f(x)$ is a real or complex valued continuous function for real numbers x in the interval $[m, n]$, then the integral can be approximated by the sum (or vice versa)

$$\sum_{i=m+1}^n f(i) = \int_m^n f(x) dx + \frac{f(n) - f(m)}{2} + \sum_{k=1}^{\lfloor \frac{p}{2} \rfloor} \frac{B_{2k}}{(2k)!} \left(f^{(2k-1)}(n) - f^{(2k-1)}(m) \right) + R_p$$



We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$= \sum_{j=0}^{n-1} \ln \left[\frac{e^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left[1 + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right] \right], \quad (3.3)$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$= \sum_{j=0}^{n-1} \left[-nV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right] \quad (3.3)$$

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THEOREM

$$-n \sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = -n^2 I_Q[\mu_Q] + n U_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) + \mathcal{O}(n^{-2}),$$

where

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$-2 U_{\mu_Q}(0) := \int_{r_0}^{r_1} \ln(s) (s q'(s))' ds = q(r_1) - q(r_0) + 2 \ln(r_1).$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned}
 &= -n^2 I_Q[\mu_Q] + n U_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) \\
 &\quad + \sum_{j=0}^{n-1} \left[\frac{1}{2} \left(\ln(2\pi r_{\tau(j)}^2) - \ln n - \ln \left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4} \right) \right) + \frac{1}{n} \mathfrak{B}(r_{\tau(j)}) + \mathcal{O}(n^{-2}) \right] \quad (3.3)
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 \end{aligned}$$

THEOREM

Asymptotically on n ,

$$\begin{aligned}
 \sum_{j=0}^{n-1} \ln V_{\tau(j)}^{(2)}(r_{\tau(j)}) &= nE_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) + \mathcal{O}(n^{-1}), \\
 \sum_{j=0}^{n-1} \ln \left(2\pi r_{\tau(j)}^2 \right) &= n \ln(2\pi) + 2 \left(-nU_{\mu_Q}(0) - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) + \mathcal{O}(n^{-1}) \right), \\
 \frac{1}{n} \sum_{j=0}^{n-1} \mathfrak{B}(r_{\tau(j)}) &= \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) s V^{(2)}(s) ds + \mathcal{O}(n^{-2})
 \end{aligned}$$

where

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds.$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j &= -n^2 l_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6} \ln \left(\frac{r_0}{r_1} \right) - \frac{n}{2} \ln(n) - \frac{1}{2} \left(nE_Q[\mu_Q] - \frac{1}{2} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \right) \\ &\quad + \frac{1}{2} \left(n \ln(2\pi) + 2 \left(-nU_{\mu_Q}(0) - \frac{1}{2} \ln \left(\frac{r_1}{r_0} \right) \right) \right) + \frac{1}{4} \int_0^{r_1} \mathfrak{B}(s) s V^{(2)}(s) ds + \mathcal{O}(n^{-1}) \quad (3.3) \end{aligned}$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j = & -n^2 l_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ & - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(n^{-1}). \quad (3.3) \end{aligned}$$

We return to $\sum_{j=0}^{n-1} \ln h_j$ using the previous asymptotic

$$\begin{aligned} \sum_{j=0}^{n-1} \ln h_j = & -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln(n) + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) - \frac{2}{3} \ln \left(\frac{r_0}{r_1} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) \\ & - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr + \mathcal{O}(n^{-1}). \end{aligned} \quad (3.3)$$

THEOREM

Asymptotically on n ,

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2} \ln n + n \left(\frac{\ln(2\pi)}{2} - \frac{1}{2} E_Q[\mu_Q] \right) + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1}) \quad (3.4)$$

where

$$F_V[A_{[r_0, r_1]}] := -\frac{2}{3} \ln \left(\frac{r_1}{r_0} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr,$$

THEOREM

For the annulus case, where $r_0 > 0$, we have the following expression for the free energy asymptotically on n given by

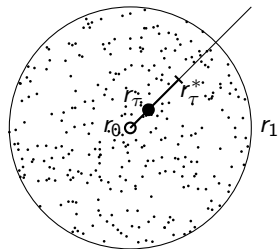
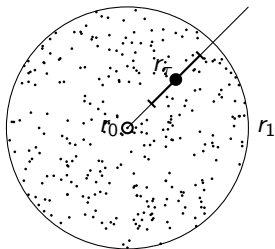
$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{n}{2} (\ln(2\pi) - E_Q[\mu_Q] - 2) + \frac{n}{2} \ln(n) + \frac{\ln(2\pi)}{2} + \frac{1}{2} \ln(n) + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1})$$

where

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds.$$

$$F_V[A_{[r_0, r_1]}] := -\frac{2}{3} \ln \left(\frac{r_1}{r_0} \right) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)} \right) - \frac{1}{32} \left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} \right) - \frac{1}{12} \int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr,$$



THEOREM

For the disk case, where $r_0 > 0$, we have the following, asymptotically on n , for the free energy

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) - \frac{5}{12} \ln n \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + \mathcal{O}(n^{-1/12} (\ln n)^3).$$

where

$$I_Q[\mu_Q] := \left[\frac{1}{4} \int_{r_0}^{r_1} q(s) s V^{(2)}(s) ds - \ln(r_1) + \frac{1}{2} q(r_1) \right],$$

$$E_Q[\mu_Q] := \frac{1}{2} \int_{r_0}^{r_1} \ln \left(\frac{V^{(2)}(s)}{4} \right) (sq'(s))' ds,$$

$$F_V[D_{r_1}] := \frac{1}{6} \ln \left(\frac{4}{V_0^{(2)}(0)} \right) - \frac{1}{3} \ln(r_1) + \frac{5}{32} \ln \left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)} \right) - \frac{1}{32} \frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12} \int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r dr.$$

For the annulus,

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) - \frac{5}{12} \ln n + \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + \mathcal{O}(n^{-1/12} (\ln n)^3).$$

while for the disk

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2} (\ln(2\pi) - 2 - E_Q[\mu_Q]) + \frac{1}{2} \ln(n) + \frac{\ln(2\pi)}{2} + F_V[A_{[r_0, r_1]}] + \mathcal{O}(n^{-1}).$$

Obrigado!

Projeto realizado com auxílio da FAPESP sob processo # 2024/06997-1.

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João V. A. Pimenta
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